Federica Domecq Lacroze

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EDUCATION

Wake Forest University — Winston-Salem, NC
M.S. in Statistics, Teaching Assistantship — Aug. 2026 – May 2028

Pomona College — Claremont, CA
B.A. Mathematics & Computer Science | Magna Cum Laude, GPA: 3.978/4.00 — Aug. 2022 – May 2026
Phi Beta Kappa & Sigma Xi Honour Societies

Honours: Bruce Jay Levy Senior Prize in Mathematics; Distinction in Mathematics Senior Exercise; Multi-semester Pomona College Scholar; Bernard Charnwut Chan Leadership & Service Award; Sheri Sangji Memorial Award

Relevant coursework:

- **Statistics:** Computational Statistics; Methods in Biostatistics; Statistical Theory; Time Series; Linear Models; Statistical Learning; Operations Research; Real Analysis I & II; Probability; Combinatorial Mathematics; Linear Algebra
- **CS:** Machine Learning; Data Visualisation; Advanced Algorithms; Computer Systems; Data Structures & Advanced Programming; Languages & Theory; Discrete Math & Functional Programming

EXPERIENCE

Pomona College — Claremont, CA

Kenneth Cooke Research Fellow – "On Par Golf: Leveraging GPR for Perfect Golf" — May 2025 – Aug. 2025

- **National USRESP Competition Winner (1st Prize):** Lead independent research applying spatial data analysis and machine learning to optimise golf course strategy; presented findings at eCOTS 2026 and currently developing a manuscript for peer review.
- **Spatial Data Pipeline:** Digitised real-world golf courses into a 280-point coordinate grid using QGIS and `Shapely`, engineering a custom pipeline to simulate terrain characteristics and spatial difficulty.
- **Advanced Predictive Modeling:** Built continuous risk-surface models using Gaussian Process Regression (`GPyTorch`) to predict outcomes, fusing simulated and real-world data via a multi-fidelity framework to systematically correct baseline biases.
- **Simulation & Optimization:** Executed high-volume Monte Carlo simulations to evaluate expected outcomes across two different loss functions, validating model stability and confirming existing sports analytics literature.

Senior Thesis (Distinction) in Mathematics — Jan. 2024 – May 2026

- **Mathematical Modelling & Proofs:** Awarded Distinction for developing two independent proofs (combinatorial and topological) expanding on Arrow's Impossibility Theorem to model moral compromise in voting systems.
- **Global Empirical Analysis:** Designed and executed a data-driven analysis of electoral systems and democratic-freedom indices across 170+ countries to ground theoretical models in real-world geopolitical data.

Statistics/Data Science & Probability Teaching Assistant — Jan. 2024 – May 2026

- Teaching Assistant for Probability (3×), Introductory Statistics (2×), Data Science (1×) over 5 semesters; led sessions on hypothesis testing, estimation, regression, ANOVA, and simulation.
- Mentored students in `R`/`Tidyverse` data pipelines, personalised code reviews, and environment setups (e.g., RStudio/Positron) to build foundational data problem-solving skills.

Summer Research, Professor Assistantship — May 2024 – Aug. 2024

- Applied sigmoidal & impulse models (`Sicegar`, `ImpulseDE2`, `DESeq2`) to gene expression data; parallelised high-volume gene expression simulations across HPC clusters using job schedulers (like `Slurm`), significantly reducing computational runtime

Resident Advisor — Dec. 2024 – May 2025

- Supervised 100+ students in a 400-resident zone; planned 5+ community programs per semester; resolved conflicts, responded to crises, and liaised with campus staff on student welfare.

DoctorSIM — Madrid, Spain

Marketing & Administrative Intern — Dec. 2021 – Jul. 2022

- Managed invoicing, international logistics (DHL), and financial record-keeping while building a 50+ client SEO database.

ACTIVITIES & LEADERSHIP

Women's Varsity Golf – *2-Year Team Captain, Pomona-Pitzer* — 2022 – 2026

- **3× NCAA National Appearances**; 2 Individual National Titles; 2× Team Conference Champions; 3× All-Conference, 2× All-Region; WGCA All-American Scholar 2023, 2024, 2025, 2026.

Co-President, AWM Claremont Chapter — 2023 – 2025 | **Math Department Liaison** — 2025 – 2026

SKILLS & LANGUAGES

Programming: `Python` (GPyTorch, Pandas, Shapely); `R` (Tidyverse); `SQL`; `JavaScript` (D3); `Haskell`; LaTeX; `HTML/CSS`
Tools: Git; QGIS; Excel; HPC & Parallel Processing; LLM-assisted coding & documentation
Languages: Fluent English, Fluent Spanish, Intermediate French, Intermediate Portuguese